AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 6 · STUDENT LESSON

Label Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Write the rule. Match the IDs. Keep uncertainty visible.

Measure annotator agreement without confusing it with correctness.

Keep one habit: Lane rule → paired count → leaf contrast → complete twenty-card lab → independent P–T task.

MAIN LESSON · SECTION 1

Define the target before asking people to label it

> Agreement describes matching decisions, not proof that either decision is correct.

Learning goals

I can write an observable **label rule** with boundaries and a review option.

I can measure **agreement** using matched items and an explicit denominator.

I can investigate dis**agreement** and record justified label decisions.

Recall inspection and missingness

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.
- Why should an unknown duration remain different from zero in a dataset card?

End goal

By the end, I can explain label quality using a traced example and a stated limitation.

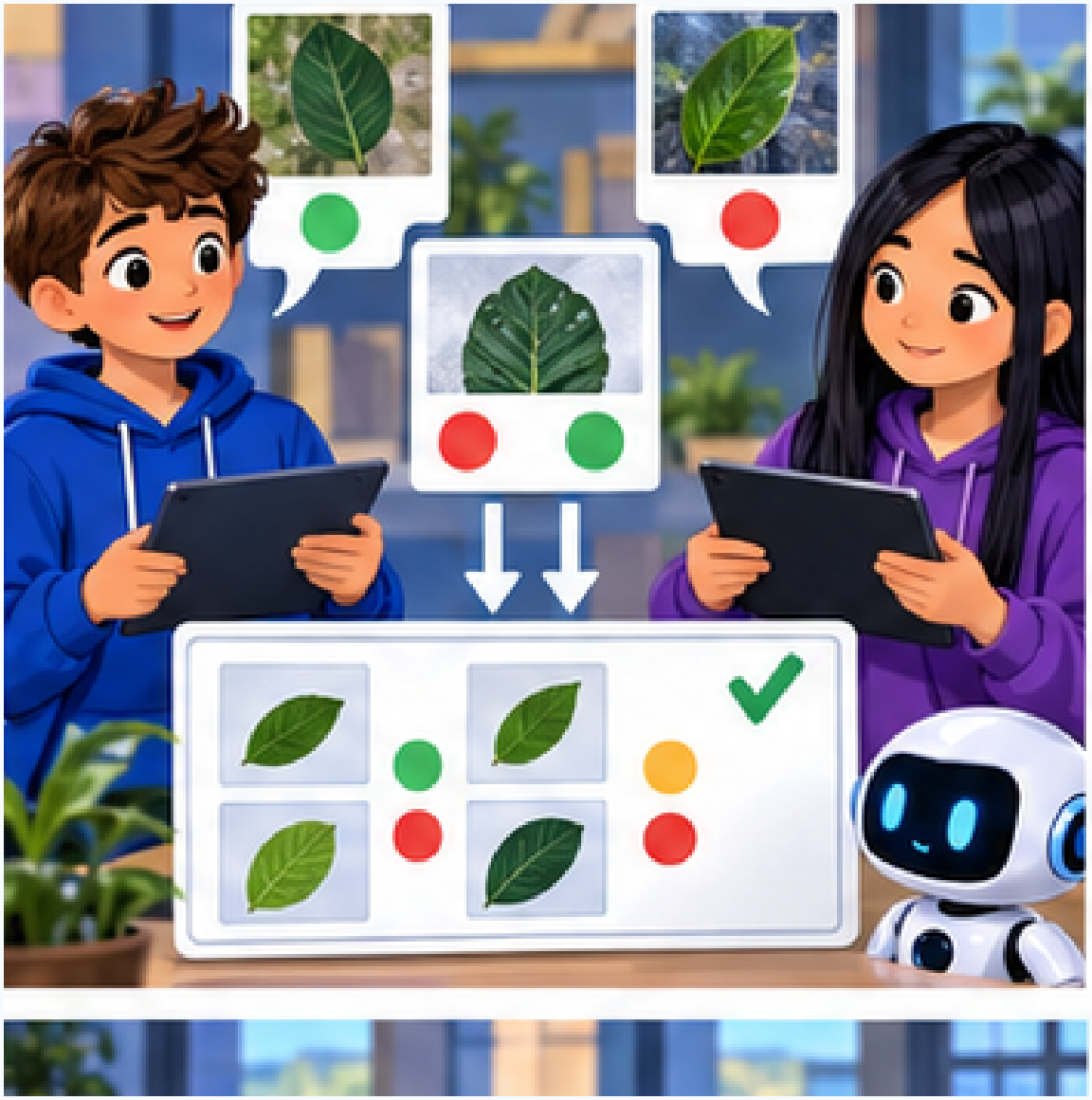
▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Ambiguous definitions and inconsistent **annotation** can make the learning target unreliable; annotator agreement alone is not a universal numerical ceiling on model performance.



LEARNING TARGET Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Use an observable boundary rule and a review route

> The corridor descriptions are classroom exercises, not robot safety guidance.

Purpose and exact rule v1

A model learns from examples with recorded targets. If two people interpret a target differently, their labels can mix different tasks.

This week you will write a **label rule**, compare independent labels and keep a record of disputed cases. We use invented corridor descriptions, not real images or a system for controlling a robot.

The classroom task is to label whether a marked practice lane is blocked in a supplied description.

Rule v1 says blocked means an object touches or crosses either lane boundary or lies inside the lane; clear means all objects are outside both boundaries and the entire lane is visible.

Use review if visibility or position is insufficient to decide. Review is a workflow status for a person to inspect, not a third physical condition of the lane.

Typical, touching and hidden examples

Description	Rule v1 result	Reason
A box lies inside a fully visible lane	blocked	Inside satisfies the rule
A wheel touches the boundary line	blocked	Touching is explicitly included
All objects are outside and the lane is fully visible	clear	Both clear conditions hold
A curtain hides part of the lane	review	Hidden space cannot be checked

Keep the use bounded

These are exercise definitions, not real robot safety rules. Do not infer that a lane is physically safe from this label. The label describes only the supplied **evidence** under this version of the rule.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use label quality to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Count matched labels on matched item IDs

> Keep original annotation pairs separate from adjudicated decisions.

Earlier vague instruction and supplied pairs

Two **annotator**s used an earlier vague instruction, mark an obstructed lane, on four invented items. Their labels are shown in the code. A and D match; B and C differ.

Compare the same item IDs, not merely the first row in two differently ordered files.

```
pairs = [("A", "blocked", "blocked"),
         ("B", "clear", "blocked"),
         ("C", "review", "clear"),
         ("D", "clear", "clear")]
agree = 0
for item_id, first, second in pairs:
    if first == second:
        agree = agree + 1
print(agree, len(pairs))
```

Trace agreement and denominator

The **output** is 2 4: two matching labels among four paired items, so observed exact **agreement** is 2/4 or 50 percent. Here review is included as a label value in the comparison: review versus clear is a dis**agreement**.

This simple count is not model accuracy and does not establish which labels are correct.

Follow a disagreement back to evidence

OBSERVE	INVESTIGATE	RECORD
B has two labels	Read boundary rule	Keep labels and reason
C includes review	Check visible evidence	Retain uncertainty if needed
A and D match	Check rule as well	Agreement is not proof

Preserve independent labels before discussion. Shared mistakes can produce agreement, and missing evidence can prevent a justified final label.

State pair-validation assumptions

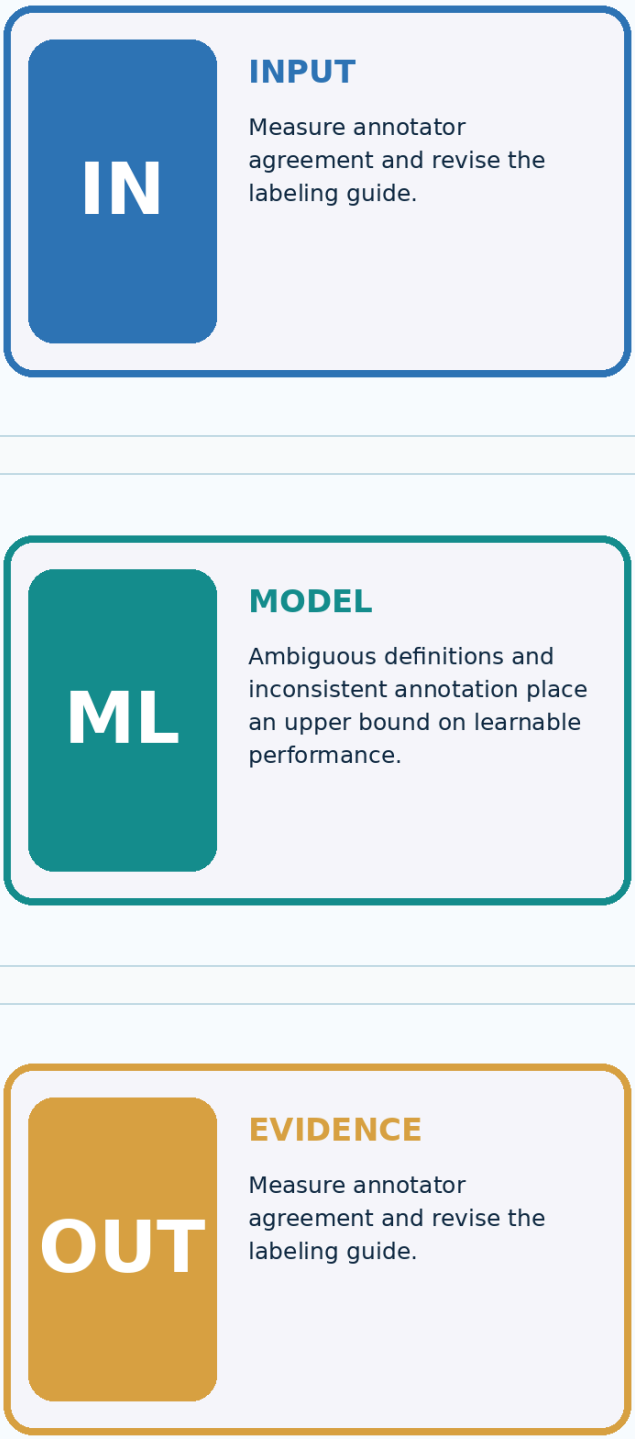
The code assumes one valid pair per unique item ID and allowed label strings. It does not validate IDs or spelling. Use a nonempty paired set before calculating a fraction; zero paired items gives no **agreement** estimate.

▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



QUESTION Measure **annotator agreement** and revise the labeling guide.

MODEL IDEA Ambiguous definitions and inconsistent **annotation** can make the learning target unreliable; annotator agreement alone is not a universal numerical ceiling on model performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Two **annotator**s label 20 leaves and agree on 16. Observed **agreement** = 16 ÷ 20 = **80%**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

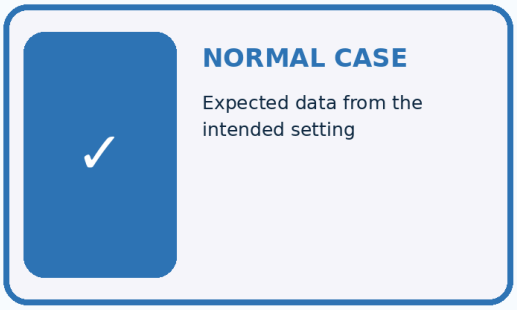
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

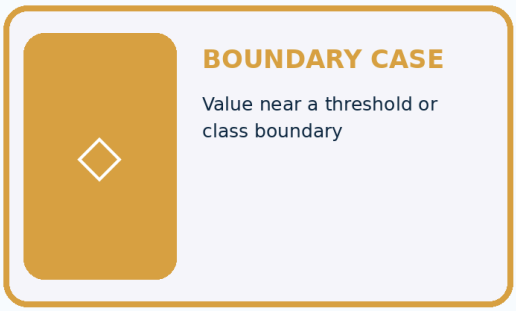
Compare a clear decision with insufficient visible evidence

> Rule clarity can resolve a boundary; voting cannot reveal a hidden region.

Four evaluation conditions



NORMAL Expected data and expected behavior.

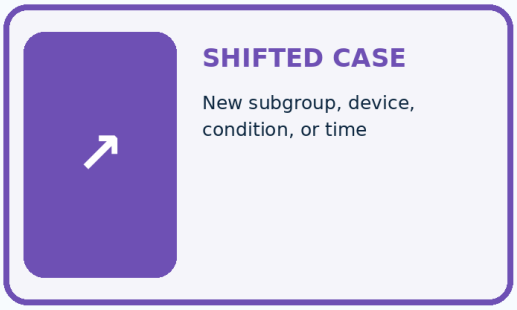


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Z receives a new fully visible image with no brown spots.

▼ **Check after predicting**

Prediction: Label the new **observation** healthy under the toy rule.

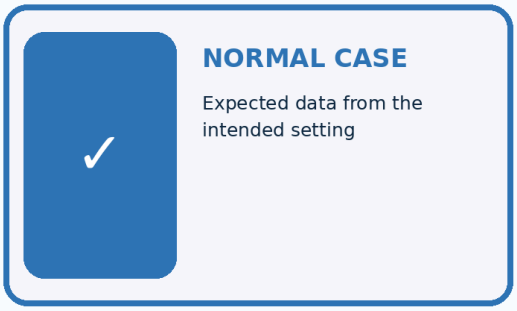
Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

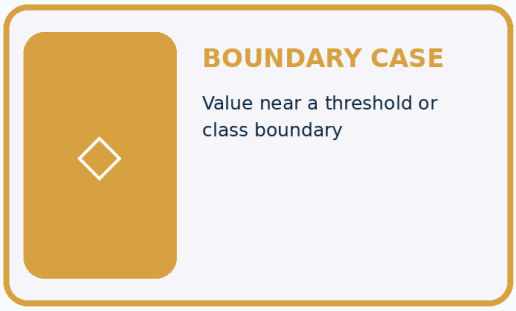
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

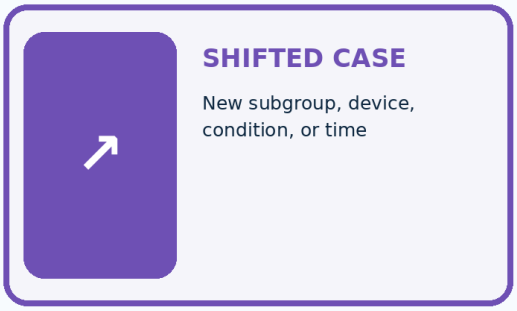
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Resolve the boundary and preserve the uncertain case

> Document what the rule and visible evidence actually justify.

Read separate leaf facts and scope

Write a **label rule** that resolves the visible boundary case.

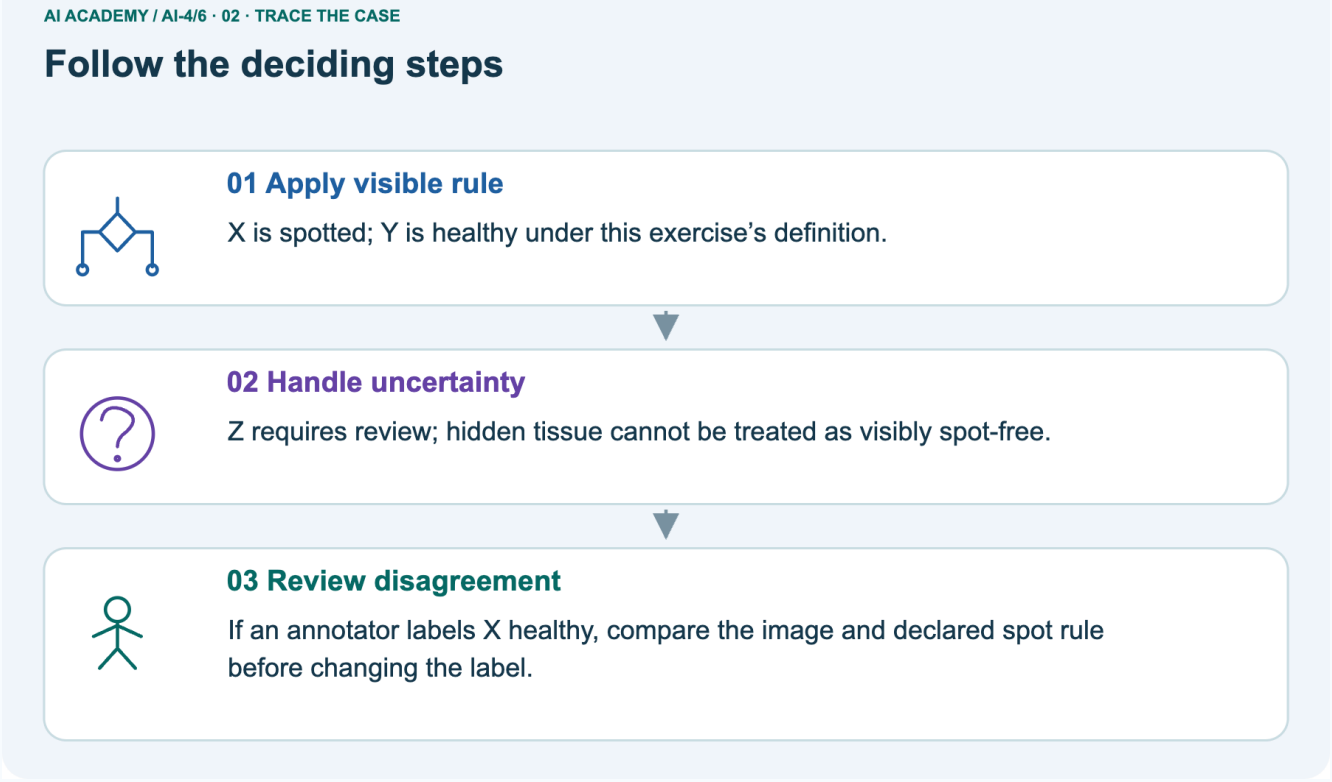
Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The leaf picture uses a separate toy rule. Healthy here means no visible brown spots on a fully visible leaf under this exercise, not actual plant health. A new fully visible image supplies new evidence; it does not make the older hidden image informative. For corridor pairs A–D,

excluding C leaves A,B,D with two original matches out of three; reviewed labels do not replace the original comparison.

Trace labels, review and limits



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Apply visible rule	X is spotted; Y is healthy under this exercise's definition.
2. Handle uncertainty	Z requires review; hidden tissue cannot be treated as visibly spot-free.
3. Review disagreement	If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

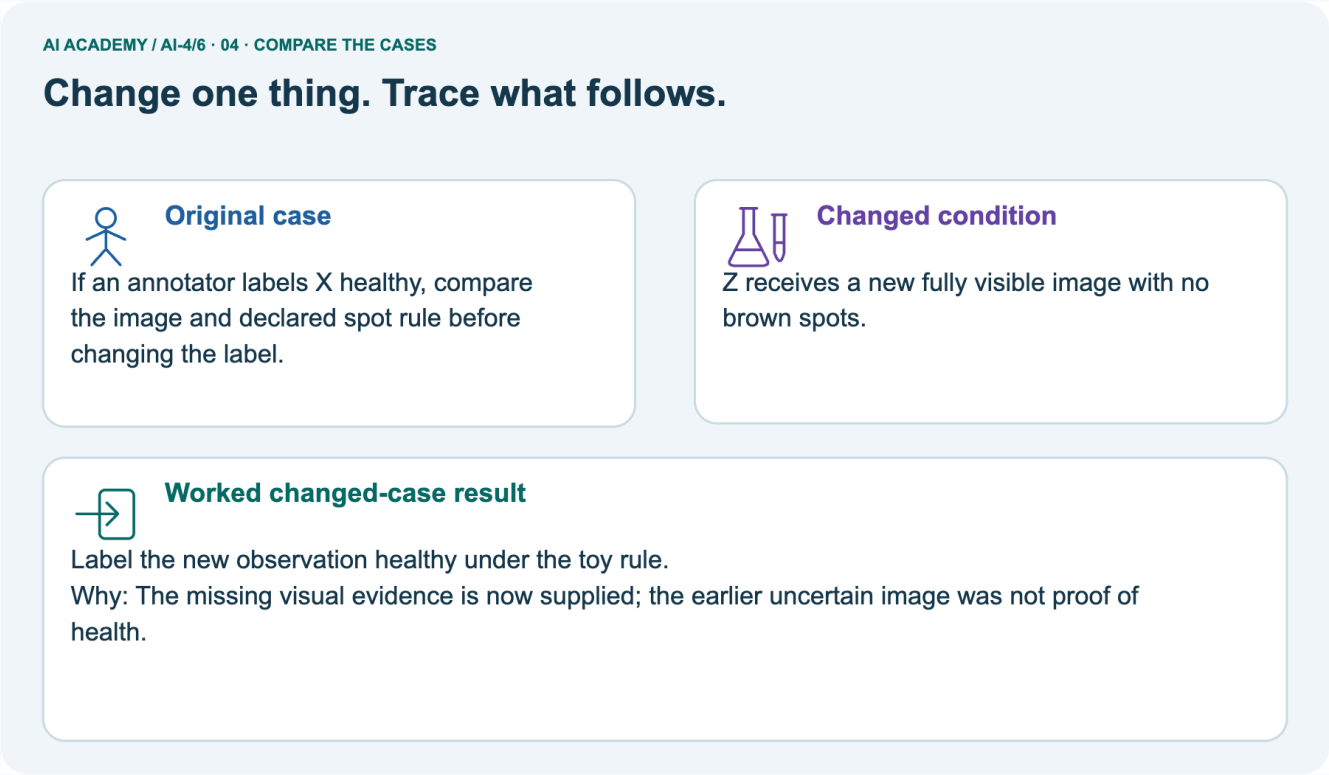
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

Predict a newly visible observation

- Change one thing: Z receives a new fully visible image with no brown spots. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why votes cannot reveal hidden evidence

- Read the labelled picture: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Review the corridor B and C decisions

For B, the full lane is visible and a wheel touches its boundary. One **annotator** treated touching as clear; the other treated it as blocked. Rule v1 explicitly includes touching, so the reviewed label is blocked.

Preserve both original labels and record the item, rule version, **evidence**, decision and reviewer. Do not silently overwrite the originals.

For C, a curtain hides part of the lane. One **annotator** guessed clear; the other requested review. Rule v1 requires full visibility for clear. The justified status remains review until sufficient **evidence** is available.

More votes cannot reveal the hidden area.

> **Using rule v1, explain the decisions for B and C. If you exclude C from the original comparison, what is agreement among the remaining pairs? Why must you still report the exclusion?**

A decision log could say: B, original clear/blocked, wheel touches line, v1, blocked after review by the teacher. For C it could say: original review/clear, lane partly hidden, v1, unresolved review.

Add the labeling method and rule version to the dataset card.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 6 • WORKED EVIDENCE

1. Two annotators label 20 leaves and agree on 16.
2. Observed agreement = 16 ÷ 20 = 80%.
3. Most disagreements concern tiny spots near the boundary.
4. Define a minimum spot rule, relabel a fixed subset, and report the new agreement.

STEP 1 Two **annotator**s label 20 leaves and agree on 16.

STEP 2 Observed **agreement** = 16 ÷ 20 = **80%**.

STEP 3 Most dis**agreement**s concern tiny spots near the boundary.

STEP 4 Define a minimum spot rule, relabel a fixed subset, and report the new **agreement**.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

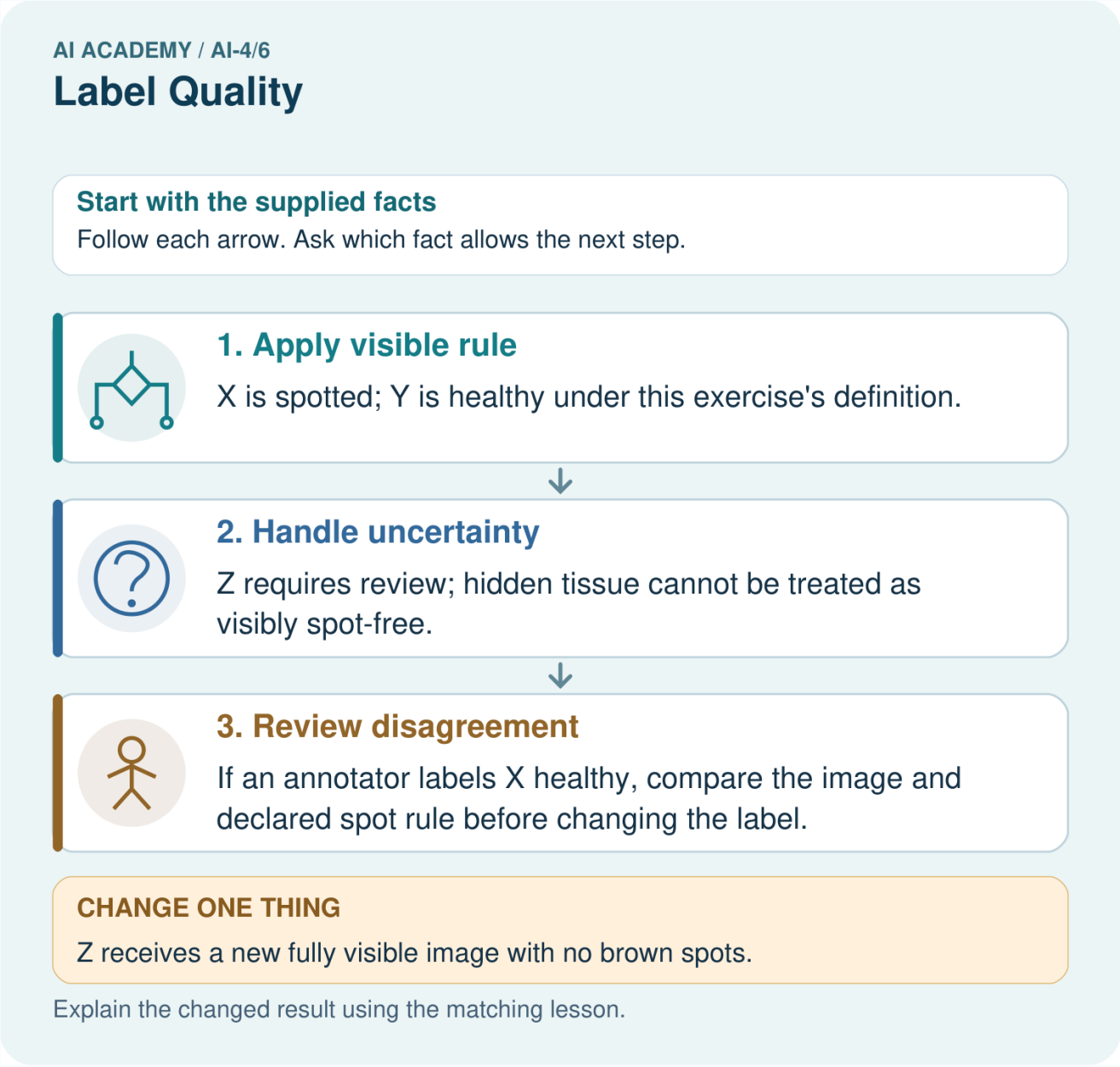
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Label Quality

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Separate agreement, correctness and rule changes

> A shared mistake can produce perfect agreement.

State count, subset and rule version

Report the paired-item count, **agreement** count, handling of review and the rule version. For the original four-item comparison, say two of four pairs match, including review statuses.

If you report a subset, name its selection rule and denominator. This tiny invented set says nothing about reliability on other scenes.

Keep practice for writing **label rule**s separate from final model testing. Resolving labels needs a documented **process**; reviewing model **prediction**s to choose convenient labels would undermine a fair evaluation. Next week examines sampling and intended use; later split-design lessons separate related examples.

Explain a shared error

> **Two annotators agree on every item but both ignore the boundary rule. What does perfect agreement establish, and what does it fail to establish? Give one evidence check to perform next.**

Challenge the claim with evidence

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE The model can learn the correct concept even when people label the same example differently.

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model can learn the correct concept even when people label the same example differently. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise the original comparison before the fresh task

> Track which rule produced which labels.

Try W2 and explain the denominator

Use workbook W2 for the connected example “Label definitions and dis**agreement**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Write and pilot an observable rule

A useful rule names the object of judgment, allowed labels, observable **evidence**, boundaries and an uncertainty route. Include a typical example and a boundary example.

Pilot the rule independently on a few new items, compare labels, discuss causes, revise the wording and label new items again. Record which rule produced each set of labels.

Do not change the rule simply to improve a score or make a model look correct. A rule revision can change the target itself.

If touching is changed from blocked to clear, document the changed task, review all affected examples consistently and keep versions separate.

Consult task **evidence** when interpreting a dis**agreement**; a majority vote alone cannot prove truth.

Check agreement-is-accuracy reasoning

- Use the worked case for Label definitions and dis**agreement**. Proposed explanation: "**Agreement** is accuracy". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain missing evidence

- Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the independent task

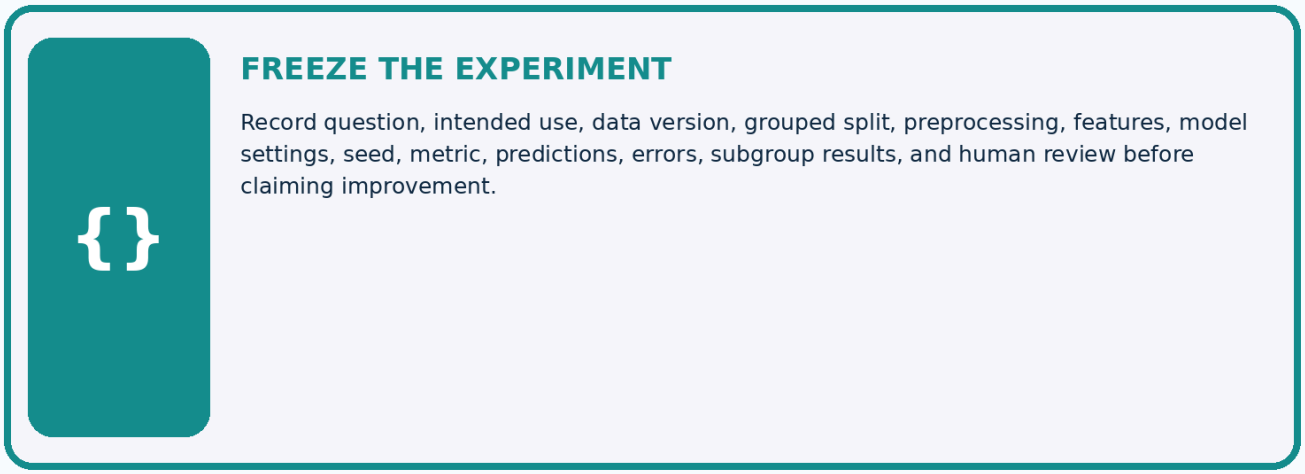
This is supported practice. Keep the separately named independent case for an attempt before feedback.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.
</details>



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Interpret agreement

Check your reasoning

Explain the picture

Run Label Lab with all twenty cards supplied

> Measure real agreement rather than inventing an annotator outcome.

Twenty complete label cards

Card ID	Complete fictional description
L01	A box lies inside the fully visible lane.
L02	A wheel touches the left boundary of a fully visible lane.
L03	Every object is outside both boundaries and the whole lane is visible.
L04	A curtain hides the middle of the lane; no object position inside can be established.
L05	A cable crosses the right boundary of a fully visible lane.
L06	A cone is outside; every other object is also outside and the lane is fully visible.
L07	An object is partly hidden and its relation to a boundary cannot be determined.
L08	A ball lies inside the fully visible lane.
L09	A wheel touches the right boundary of a fully visible lane.
L10	No objects are inside or touching either boundary; the entire lane is visible.
L11	A screen hides the far end; no known object inside or touching is established.
L12	A strip crosses the left boundary of a fully visible lane.
L13	All boxes lie outside both boundaries; the full lane is visible.
L14	A photo crop hides part of the lane and object positions there are unknown.
L15	A toy lies inside a fully visible lane.
L16	A marker touches the left boundary of a fully visible lane.

Card ID	Complete fictional description
L17	The entire lane is visible and every marker is outside both boundaries.
L18	Glare prevents judging part of the lane; no blocking object is otherwise established.
L19	A rope crosses both boundaries of a fully visible lane.
L20	The whole lane is visible and all chairs are outside both boundaries.

Normal, boundary, missing and changed-description probes

Use L01 as a normal inside example, L02 as a touching boundary, L04 as missing visibility, and L03 as a clear example. Changed wording probe: replace L03's full visibility with a hidden middle region and unknown contents. Predict which clause changes the status. This is a new description, not evidence that the old observation was wrong.

Independent labeling, review and documented repeat

1. Two learners independently apply v1 to all twenty supplied descriptions, without seeing each other's answers. Record actual pairs and matching count/20 including review. Group disagreements by boundary interpretation, visibility or recording error. Discuss evidence, write one clarification with a new version if needed, then independently repeat these same twenty cards and report that it is a repeated practice set.

   No improved count is promised; a changed rule can change the target. Retain both rounds and rule versions. Repeated-card agreement is not unseen generalization or model accuracy. Keep the separate five-item independent task for an unaided first attempt.

Record actual results or not run

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____

> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

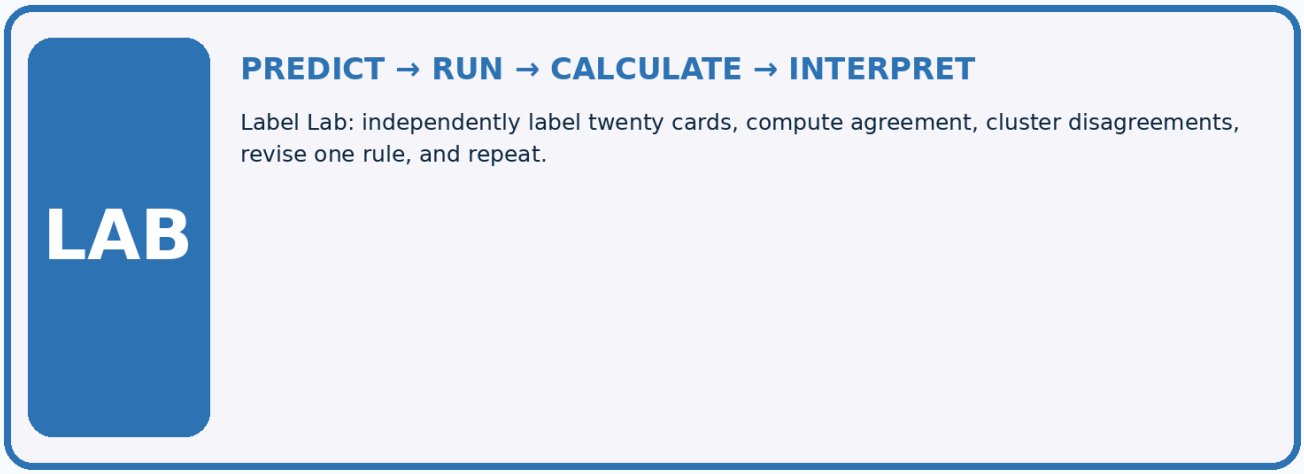
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>



1. Write the hypothesis and expected **evidence** before running.

2. Label Lab: independently label twenty cards, compute **agreement**, cluster dis**agreement**s, revise one rule, and repeat.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare and review five new annotation pairs

> Keep original agreement and reviewed decisions distinct.

Complete selected pairs and contract

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear.

P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has an object inside. Count **agreement** including review. Apply v1 to all five.

Then report **agreement** on the original pairs excluding any review status, with its denominator and limitation.

For this task, “original pairs” means the supplied P–T annotation pairs, not A–D from the worked example. Use v1: inside or touching either boundary means blocked; fully visible with all objects outside means clear; insufficient evidence means review. Compare the original annotations for agreement; keep your reviewed v1 decisions in a separate column.

When excluding review, omit pairs where either original annotation is review, and state the remaining denominator.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply the process independently

Week 6 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better **process**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute **agreement** and explain why it does not establish correct labels.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a label guide and decision history with the project

> Make uncertainty visible before a model learns from the targets.

Project guide and evidence

THIS WEEK'S CHECKPOINT Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

Save the allowed labels, observable rule, boundary examples, uncertainty route, annotation instructions and version. Retain original labels, item IDs, independent versus discussed status and review decisions with reasons. Report paired count and how review was included. A classroom spotting label is not disease diagnosis, and label agreement is not a model evaluation.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits, review and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish by explaining agreement and its limit

> Name a boundary and the evidence needed to resolve uncertainty.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Write an observable **label rule** with boundaries and a review option

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
annotation	human assignment of a label	
label rule	observable criterion for assigning a class	
agreement	fraction of labels that match	
adjudication	resolve disagreements using evidence	

Closer-explanation terms

Word	Meaning
Label rule	A written definition for assigning an outcome to an example
Annotator	A person who assigns labels using a stated rule
Agreement	How often annotators assign the same label to the same items

Word	Meaning
Adjudication	Reviewing evidence and rules to resolve a disputed label or retain uncertainty

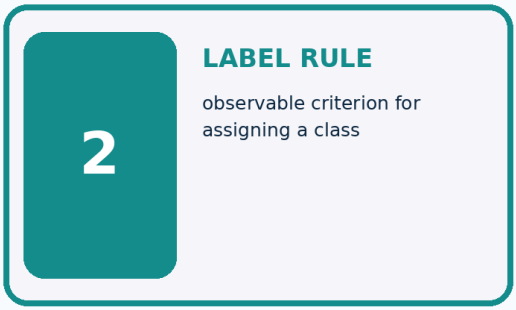
▼ Earlier wording, alternatives and picture transcripts

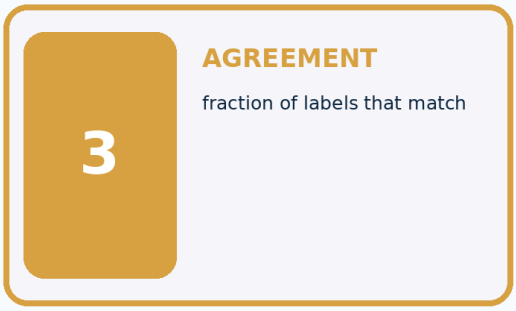
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR The model can learn the correct concept even when people label the same example differently.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **annotation**, **label rule**, **agreement**, **adjudication**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE The model can learn the correct concept even when people label the same example differently.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

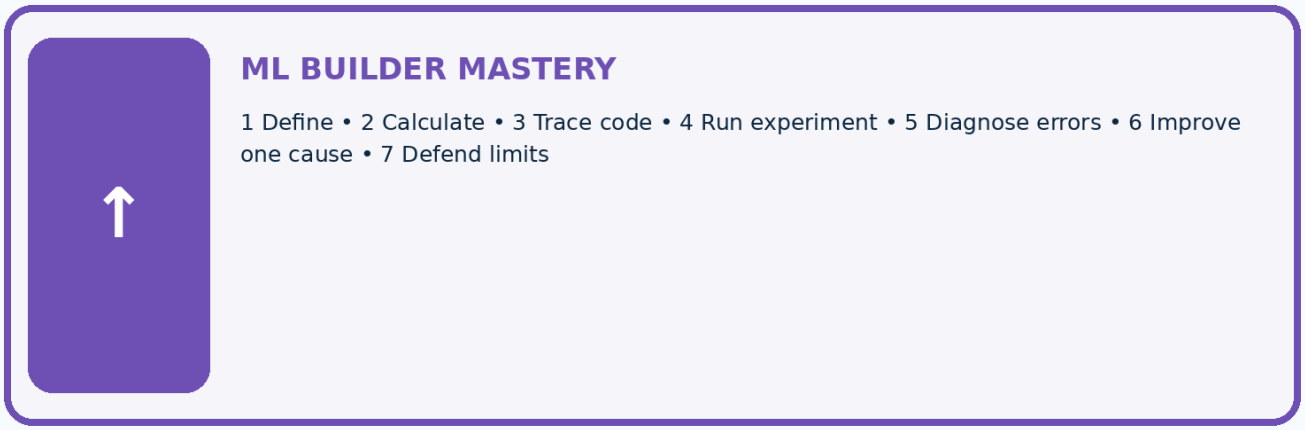
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the five new P–T annotation pairs for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.
- Independent **evidence** and defense:

- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Measure **annotator agreement** and revise the labeling guide.

NEXT BRIDGE Carry the experiment **evidence** into: Sampling and Intended Use.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 13

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

Evidence or stage	Value or result
Annotator agreement	7/10
Disagreement	3/10

▼ Earlier wording, alternatives and picture transcripts

Week 6 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Agreement is useful evidence, not truth

Two **annotator**s disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Define spotting operationally, inspect dis**agreement**, use appropriate **adjudication** and document ambiguity. **Agreement** is 7/10, but **agreement** is not proof of truth.

▼ **Reference · Earlier route checklist**

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Write an observable label rule with boundaries and a review option	Explain the deciding step, show a trace or test, and state one limit.
Measure agreement using matched items and an explicit denominator	Explain the deciding step, show a trace or test, and state one limit.
Investigate disagreement and record justified label decisions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: Write a **label rule** that resolves the visible boundary case.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Write a **label rule** that resolves the visible boundary case.
- Which labelled step matches this detail? If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.
- 1. Apply visible rule
- 2. Review dis**agreement**
- 3. Handle uncertainty
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Review dis**agreement** If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Read the picture in order.

If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Z receives a new fully visible image with no brown spots. Which conclusion is supported by the supplied facts?
- 1. Label the new **observation** healthy under the toy rule.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Label the new **observation** healthy under the toy rule. The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Repair and keep the record

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

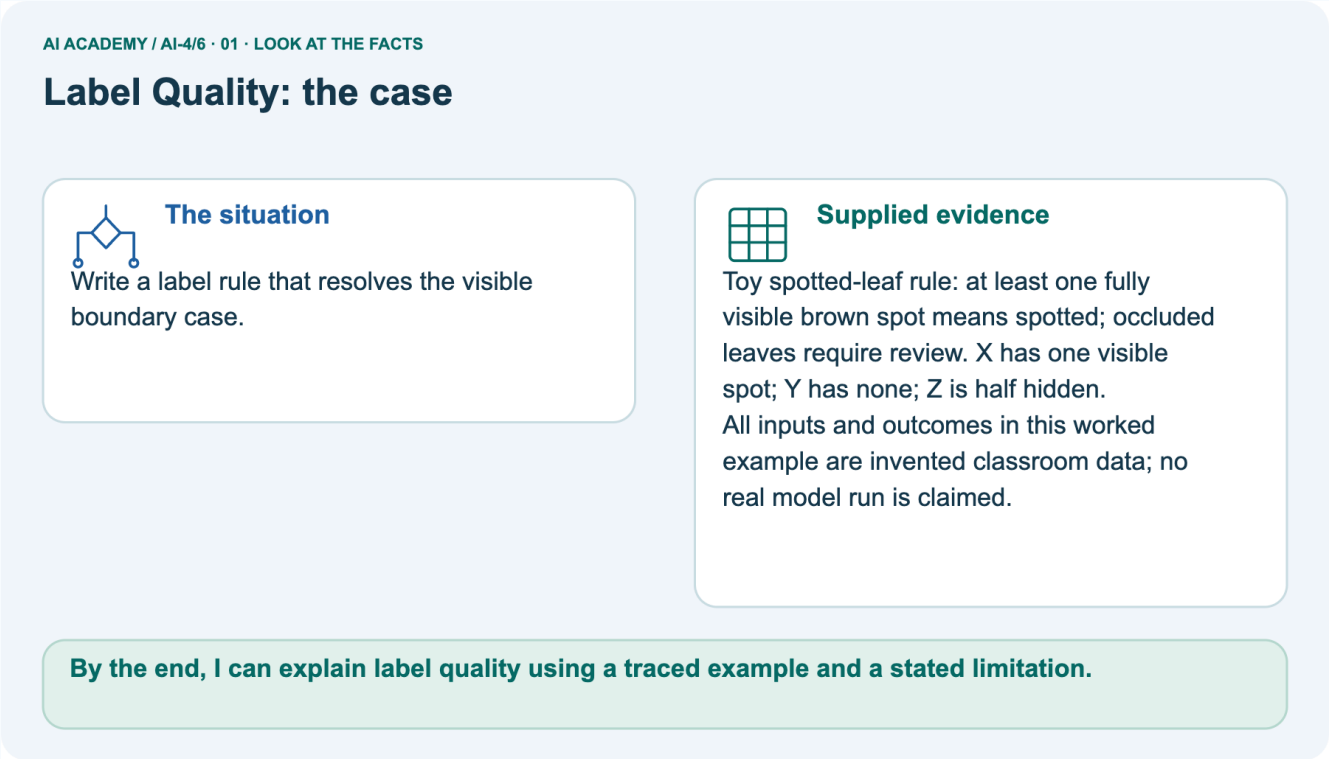
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Label Quality: the case



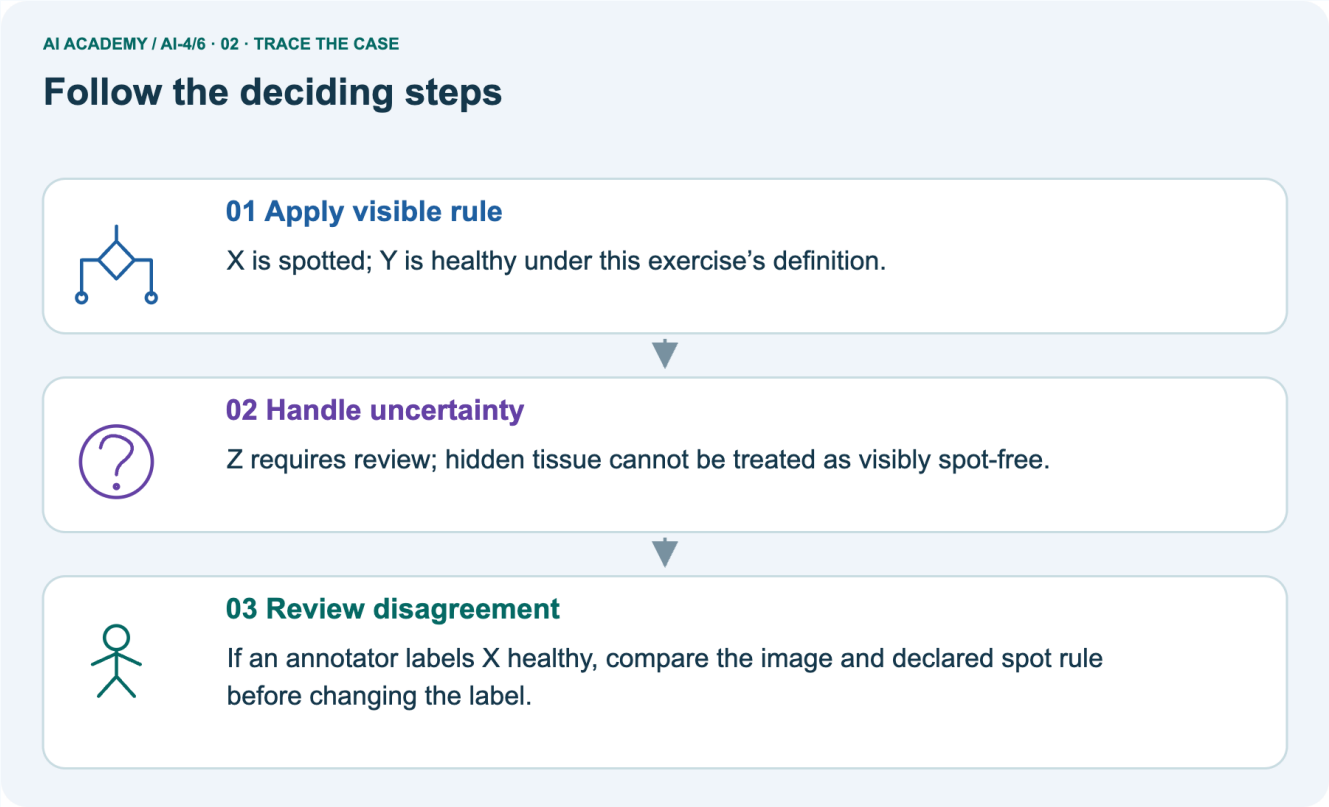
Follow the labelled picture with the matching explanation. Open the vector diagram

Write a **label rule** that resolves the visible boundary case. Supplied facts: Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review.

X has one visible spot; Y has none; Z is half hidden. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply visible rule: X is spotted; Y is healthy under this exercise's definition. → Handle uncertainty: Z requires review; hidden tissue cannot be treated as visibly spot-free. → Review dis**agreement**: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow a disagreement back to evidence



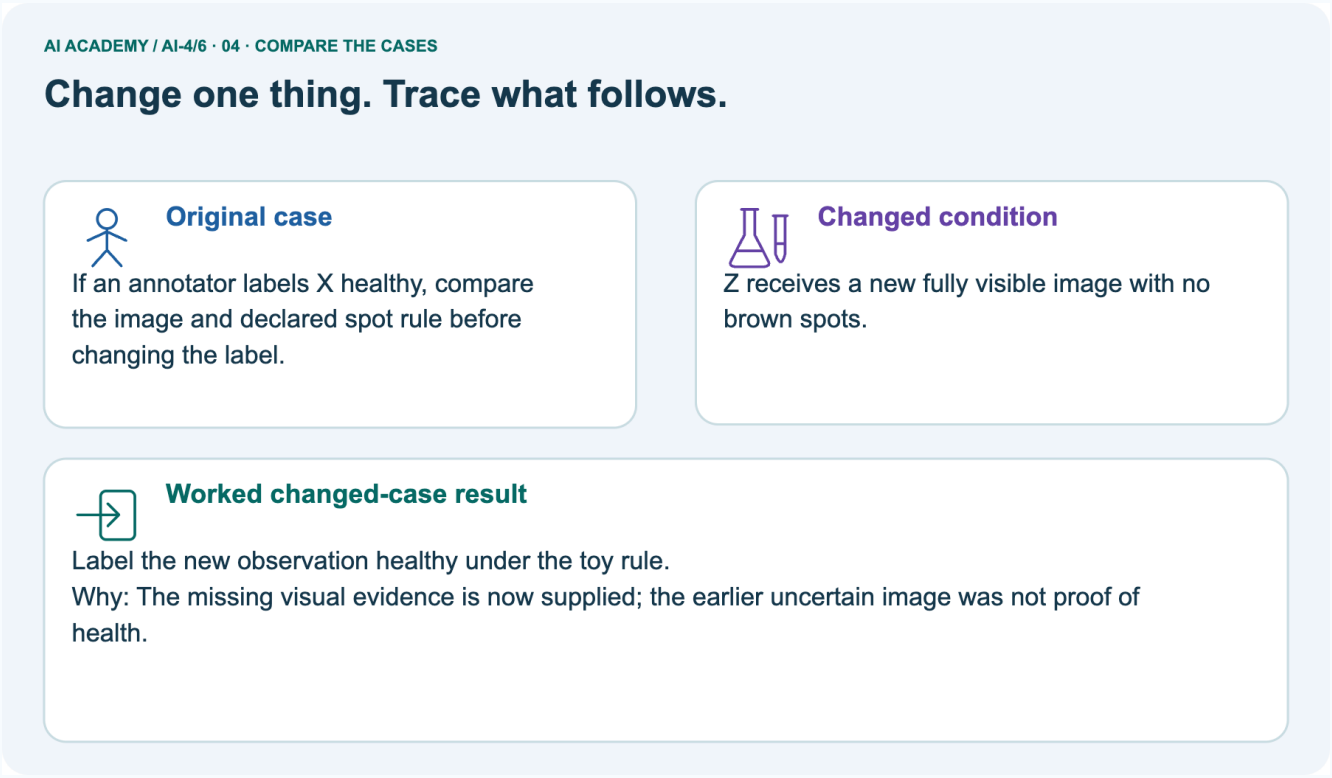
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow a dis**agreement** back to **evidence**. B has two labels | Read boundary rule | Keep labels and reason. C includes review | Check visible **evidence** | Retain uncertainty if needed.

A and D match | Check rule as well | **Agreement** is not proof

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



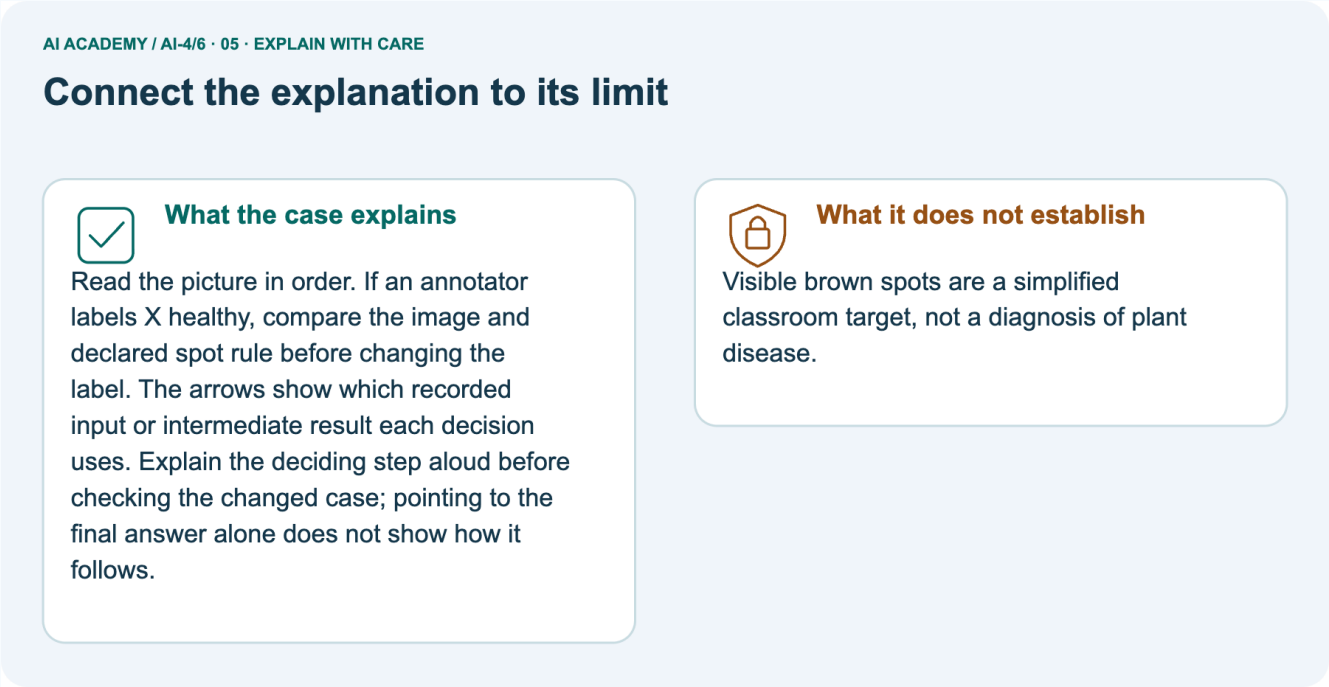
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. Changed condition: Z receives a new fully visible image with no brown spots.

Expected result: Label the new **observation** healthy under the toy rule. Reason: The missing visual **evidence** is now supplied; the earlier uncertain image was not proof of health.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



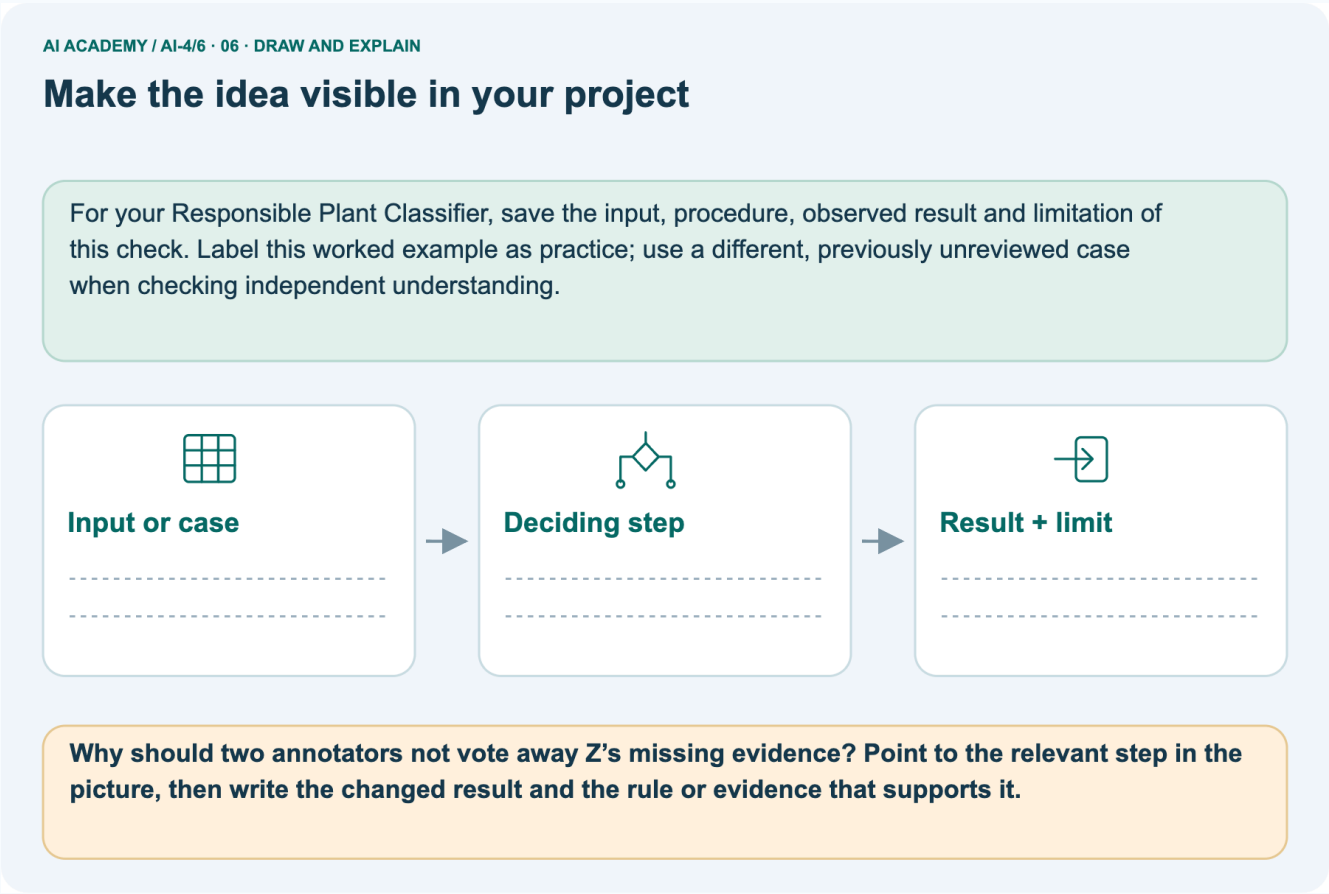
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. If an **annotator** labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should two **annotator**s not vote away Z's missing **evidence**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Label Quality: the case

02. Follow the deciding steps

03. Follow a disagreement back to evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.